

ADHIL KRISHNA G

Software Engineer (Flutter)

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Summary

Software Engineer with 4+ years building scalable, high-performance cross-platform healthcare applications using Flutter and Dart. Strong foundation in Data Structures, Algorithms, Object-Oriented Programming, and System Design principles. Proven expertise in Clean Architecture, MVVM, SOLID principles, performance optimization, RESTful API integration, state management, and Agile/Scrum methodologies. Experienced in delivering robust, production-grade applications with App Store/Play Store deployments, measurable business impact, and healthcare compliance.

Skills

Programming Languages: Dart, Python, SQL

Frameworks & SDKs: Flutter, Firebase

State Management: Bloc, Provider, Riverpod

Architecture & Concepts: Clean Architecture, MVVM, SOLID Principles, Object-Oriented Programming (OOP), Data Structures, Algorithms, System Design

UI/UX & Design: Material Design, Animations, Responsive Design, Pixel-Perfect UI

Backend & APIs: RESTful APIs, JSON, API Integration

Databases: SQLite, Hive, Drift, SQL, Firebase Firestore

Testing & Quality: Unit Testing, Widget Testing, Integration Testing, Debugging, Performance Optimization

DevOps & Tools: Git, Jira, Postman, CI/CD Pipelines, Android Studio, Visual Studio Code, Flutter DevTools

Cloud & Platforms: Firebase (Auth, Cloud Messaging, Storage), Cloud-based Application Deployment, Play Store Publishing

Experience

Software Engineer

March 2023 – Present

Glitz IT Solutions, Kerala, India

- Designed and developed scalable Flutter application architecture for a hospital management system using Clean Architecture and MVVM, improving operational efficiency by 20 percent.
- Applied data structures and enhanced API interaction patterns to reduce redundant network calls, achieving a 75 percent reduction in load time.
- Integrated secure payment gateways and third-party video communication services, enabling 500+ remote consultations per month.
- Engineered document upload, PDF processing, and medical image (DICOM) viewing modules, reducing manual administrative effort by 60 percent.
- Implemented speech-to-text AI solutions for automated clinical summaries, reducing documentation time by approximately 30 percent.
- Optimized application startup time from 7.3 seconds to 1.8 seconds using caching strategies, lifecycle management, and performance profiling.
- Improved list and grid rendering performance from 15 seconds to under 1 second using Sliver widgets, lazy loading, and animation optimizations.
- Designed fuzzy search algorithms using local databases, increasing data retrieval accuracy by 40 percent.
- Maintained REST API-driven clinical policy applications with strict adherence to client and healthcare compliance requirements, including secure data handling.
- Led sprint planning, task tracking, and peer code reviews using Jira in Agile/Scrum environment, ensuring 95 percent on-time delivery.
- Collaborated with cross-functional teams for end-to-end feature ownership, including unit/widget/integration testing and cross-platform deployment.

Support Engineer

March 2022 – February 2023

Glitz IT Solutions, Kerala, India

- Provided production support for hospital systems using SQL and SSMS, maintaining 99 percent system availability.
- Diagnosed and resolved over 200 live production issues through structured root-cause analysis.
- Implemented secure coding practices to enhance system reliability and compliance in production environments.

Projects

Hospital Management System (HMS) | *Flutter, Dart, Python, SQL, REST APIs, Payments, Video Communication, AI Automation, Jira*

- Developed a scalable end-to-end cross-platform healthcare platform for patient records, appointments, billing, and clinical reporting with pixel-perfect responsive UI.
- Applied Clean Architecture principles and optimized data structures to ensure performance, scalability, and maintainability.
- Integrated secure payments, video consultations, Firebase services, and automated clinical documentation workflows.

Clinical Policy Application | *Flutter, RESTful APIs*

- Built a dynamic content-driven cross-platform application to manage and distribute clinical policies with seamless API integrations.
- Ensured seamless updates aligned with evolving healthcare compliance requirements using state management and caching.

Chat Application | *Flutter, Firebase*

- Built a real-time cross-platform messaging application with secure authentication, Firebase Cloud Messaging for notifications, and cloud data synchronization.

Accomplishments

- Reduced application load time by 75 percent through targeted performance optimization and profiling.
- Improved local search accuracy by 40 percent using optimized query algorithms.
- Enabled 500+ monthly remote consultations through integrated video communication features.
- Achieved up to 99 percent improvement in UI rendering performance using Flutter Sliver widgets and animation techniques.
- Integrated third-party libraries for secure payments and video calls, enhancing app security and user trust in compliance-heavy environments.

Certifications

- Professional training in Python and SQL, Bangalore (2022)

Education

BTech in Electrical and Electronics Engineering

2017 – 2021

APJ Abdul Kalam Technological University, Kerala, India Self-taught in Computer Science fundamentals via online platforms, applying DSA and OOP to mobile development.